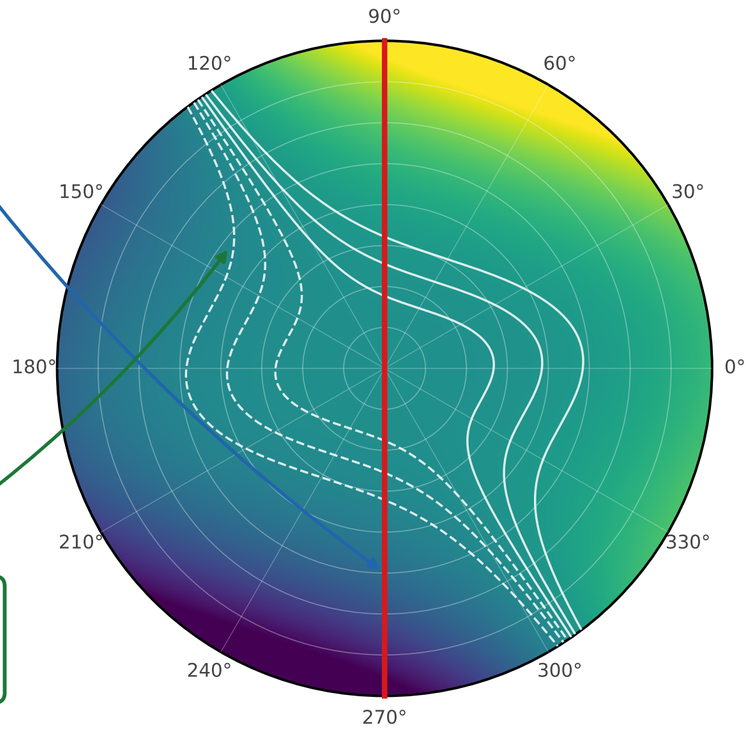
Stage 5 — The 180° half turn: polarity flip

An odd field f(v) = deg(v)·⟨v,u⟩ (weight k = 3) under v ↦ −v: the colours swap, the level curves stay. Even data (deg itself) is unchanged.

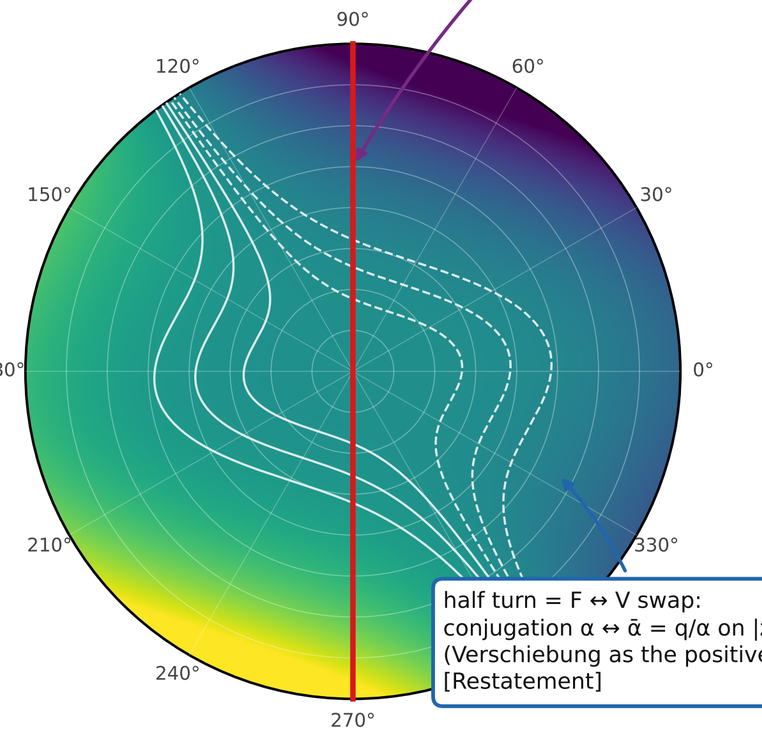
half turn S = rotation by 180°: S² = −I in SL₂(ℤ)
acts by (−1)^k on weight k (odd weights flip);
s ↦ 1 − s is the 180° turn about ½, sign = root number
(+1 for ζ, −1 for 37a1)
[Restatement]

Hardy Z(t) = e^(iϑ(t))ζ(½+it) is real and changes sign at each
odd-order on-line zero. TAA Lemma 152: ζ 38 sign changes / 38 zeros;
DH 62 / 66 on [10,120] (off-line pairs add 2 to N, no flip)
[Numerical]

before: f(v)



after the 180° turn: f(−v) = −f(v)



ghost-blind: on-line zeros flip like ζ's;
the wedge is the sign layer, not the missing surface (TAA Lemma 152, correction 8)
[Theorem]

half turn = F ↔ V swap:
conjugation α ↔ ᾱ = q/α on |z| = √q
(Verschiebung as the positive adjoint)
[Restatement]

Status tags: see legend.png. RH is not proved. J. W. McGreevy's frames flip at the 180° side (frames 5→6, 10→11; TAA Lemma 152). Per-frame colour autoscaling could fake a flip; his code is lost (TAA Lemma 152, correction 11).

After J. W. McGreevy's animation 'Flat Potential Oscillations' (Aug 2025, x.com/McGreevyMD/status/1951474097990279331). Re-rendered by Grok Bot from the analysed geometry; not J. W. McGreevy's code.